

Half-lives of serum PCB congener concentrations in environmentally exposed early adolescents.



The aim was to determine half-life of six most abundant PCB congeners in the body of early adolescents. In 304 environmentally exposed children, PCB serum concentration was determined at the age of 8 and 12years. Half-life was determined for each child assuming exponential decrease or for the whole cohort using multiple regression. Results obtained by both approaches were in agreement. PCB reuptakes corrupting half-life estimates for each child and each congener were evaluated. If one of the serum PCB concentration values fell below the level of detection (LOD) the pair was excluded and if PCB half-life value exceeded the arbitrary value of 30years. The following median half-lives in years 4.46, 10.59, 9.7, 4.7, 9.1 and 9.8 were obtained for PCB congeners 118, 138(+163), 153, 156(+171), 170 and 180, respectively. The elimination half-life values were not systematically related to PCB serum concentration at any examination age. Between half-life values, percentage of children with significant reuptakes and PCB congener abundance in serum were found significant associations. Copyright © 2010 Elsevier Ltd. All rights reserved.